

Development of a Secure IIoT Virtual Lab Environment

A Containerized Testbed for Modbus/TCP Security Research

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Abstract

The convergence of operational technology (OT) and information technology (IT) under the Industry 4.0 paradigm has expanded the attack surface of Industrial Internet of Things (IIoT) deployments. Empirical security research in this domain typically requires access to programmable logic controllers (PLCs), managed industrial switches, and licensed SCADA software—resources that are cost-prohibitive for academic institutions and individual researchers. This report documents the development of a containerized IIoT virtual lab environment built on Containernet, Docker, and Open vSwitch (OVS), executing on Ubuntu 24.04 LTS under the Windows Subsystem for Linux 2 (WSL2). A three-node topology comprising a Modbus client (sensor), a Modbus server (gateway/PLC), and an adversarial host was instantiated and validated. Connectivity benchmarks demonstrated 0% packet loss across a 50-packet ICMP stress test with sub-millisecond mean round-trip times (~0.1 ms), and Wireshark captures confirmed correct ARP and ICMP behavior at the data-link and network layers. The resulting platform provides a reproducible, low-cost foundation for adversarial testing, intrusion-detection research, and protocol-level security education. To validate the security workflow, abnormal Modbus/TCP traffic was generated using Scapy, captured with tcpdump, and analyzed in Wireshark. The resulting packet captures demonstrated repeated TCP SYN retransmissions and supported intrusion detection system (IDS) analysis of anomalous traffic behavior.

Keywords: Industrial Internet of Things; Modbus/TCP; Containernet; Open vSwitch; cyber-physical security; virtual testbed.

1. Introduction

Industry 4.0 represents the fourth industrial revolution, in which cyber-physical systems, cloud computing, and pervasive networking converge to create highly automated and data-driven manufacturing environments (Lasi et al., 2014). Central to this paradigm is the Industrial Internet of Things (IIoT), a class

of distributed systems that interconnect sensors, actuators, programmable logic controllers (PLCs), and supervisory software across plant, enterprise, and cloud boundaries. While this convergence has yielded substantial gains in operational efficiency, predictive maintenance, and supply-chain visibility, it has simultaneously dissolved the traditional air-gap between operational technology (OT) and information technology (IT), exposing legacy industrial protocols to a far broader threat landscape than they were originally designed to withstand (Stouffer et al., 2015).

Among the protocols inherited from this legacy era, Modbus/TCP remains one of the most widely deployed in industrial control. Standardized in 1979 as a serial protocol and later transposed onto TCP port 502, Modbus provides a request/response abstraction for reading and writing device registers and coils. The protocol, however, was designed under closed-network assumptions and accordingly omits authentication, integrity, and confidentiality mechanisms. As a consequence, an adversary with network adjacency to a Modbus segment can trivially enumerate slave devices, replay function codes, or inject forged write requests capable of altering physical process variables (Knapp & Langill, 2015).

Empirical investigation of these vulnerabilities, and of countermeasures such as deep packet inspection, anomaly detection, and protocol-aware firewalls, requires a representative testbed. Physical testbeds provide high fidelity but impose substantial capital and logistical overhead; commercial PLCs and managed industrial switches typically cost in the thousands of dollars per node, and their reconfiguration for repeated experimentation is time-consuming. This project addresses that gap by developing a fully software-defined IIoT virtual lab environment in which network topology, host images, traffic patterns, and attack simulations can be instantiated, destroyed, and reproduced programmatically. In addition to validating normal communication behavior, the project also demonstrates abnormal Modbus/TCP traffic generation and packet-level analysis to support intrusion detection research. The remainder of this report describes the system architecture (Section 2), the proof-of-concept implementation (Section 3), validation and attack-analysis results (Section 4), and conclusions with future work (Section 5). All additional files are available at <https://github.com/jaskirat12/Security-Solutions>

2. Methodology and System Architecture

